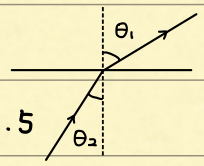
Optical Path 光程：同时间真空路程 $op := n \cdot s$

Reflection 反射　　全反射 $\theta_c = \arcsin \frac{1}{n}$　Total Reflection

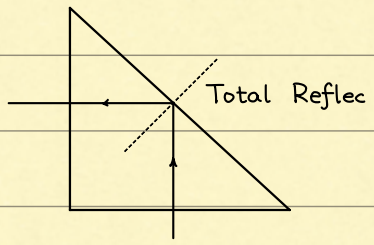
Refraction 折射

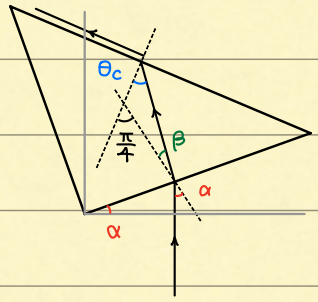
Refraction Index　$n = \frac{c}{v} = \frac{\lambda}{\lambda_0}$



$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{c}{v} = \frac{n_2}{n_1}$$ 大 小

$n_{glass} = 1.5$





$$\begin{cases} \frac{\sin\alpha}{\sin\beta} = n \\ \sin\theta_c = \frac{1}{n} \\ \beta + \theta_c = \frac{\pi}{4} \end{cases}$$

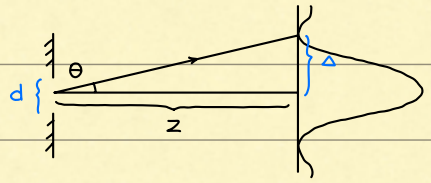
LAB 1

$$\Rightarrow \arcsin\frac{1}{n} + \arcsin\left(\frac{1}{n}\sin\alpha\right) = \frac{\pi}{4}$$

Diffraction 衍射

远场 Fraunhofer Diffraction

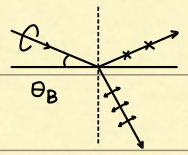
近场 Fresnel Diffraction　　$d\sin\theta = \lambda$



1st 暗条纹 Δ，$\sin\theta \approx \tan\theta = \frac{\Delta}{z}$

缝宽 aperture d　　LAB 1

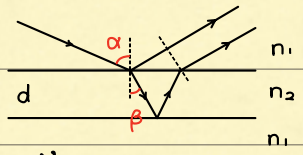
Polarization 偏振　　Scattering 散射



Brewster's angle

$$\tan\theta_B = \frac{n_1}{n_2}$$

Interference 干涉

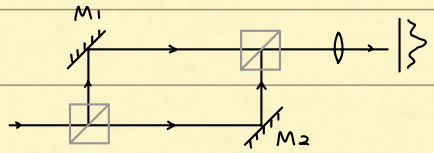


$$\frac{\sin\alpha}{\sin\beta} = \frac{n_2}{n_1}$$

$$OPD = n_2 \cdot 2\frac{d}{\cos\beta} - n_1 \cdot 2d\tan\beta\sin\alpha = k \cdot \frac{\lambda}{2}$$ 时增强

等倾干涉

e.g. 肥皂泡



Mach-Zehnder Interferometre

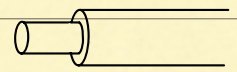
M_1. M_2 平行，则圆形

斜，则条纹 LAB 2

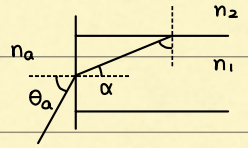
Linewidth 即 laser 的带宽，频率范围

Coherence length $L = \frac{c}{n\Delta f}$ 即波长范围

Optical Fiber 光纤



core cladding



$$\begin{cases} \cos\alpha = \frac{n_2}{n_1} \\ \frac{\sin\theta_a}{\sin\alpha} = n_1 \end{cases}$$

$$\sin\theta_a = \sqrt{n_1^2 - n_2^2} / n_a$$

NA（Numerical Apeture）数值孔径

$$NA := n_a \sin\theta_a = \sqrt{n_1^2 - n_2^2}$$

Single-Mode $M = \frac{2d}{\lambda} \cdot NA < 1$ 能量更集中

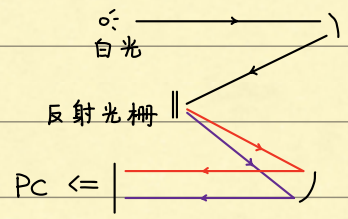
Multi-Mode 能量更分散 LAB 3

WDM：多λ传多信息

Spectroscope 光谱仪 将白光分为各色光，可用于测色

$d\sin\theta = \lambda$

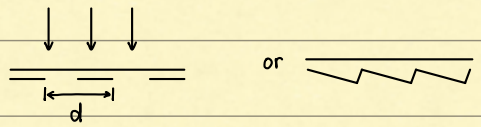
紫在中间，红在外侧



LAB 5

Grating 光栅

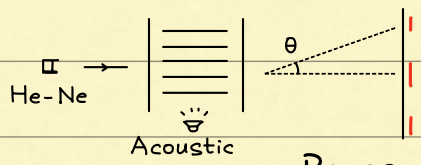
$d\sin\theta = \lambda$ d：periodicity



AOM 声光调制

声波形成光栅

可用于调激光功率



Bregg diffraction

$d \sin\theta = \lambda$

Gating periodicity

He-Ne laser wavelength

$d = v/f$

Acoustic freq

100 MHz